

第1组

1.

$$\int \frac{1}{ax+b} dx = \frac{1}{a} \int \frac{1}{ax+b} d(ax+b) = \frac{1}{a} \ln|ax+b| + C$$

2.

$$\int (ax+b)^u dx = \frac{1}{a} \int (ax+b)^u d(ax+b) = \frac{1}{a(u+1)} (ax+b)^{u+1} + C \quad u \neq -1$$

3.

$$\begin{aligned} \int \frac{x}{ax+b} dx &= \frac{1}{a} \int \frac{ax+b-b}{ax+b} dx \\ &= \frac{1}{a} \left[\int \left(1 - \frac{b}{ax+b}\right) dx \right] \\ &= \frac{1}{a} \left(\int 1 dx - \int \frac{b}{ax+b} dx \right) \\ &= \frac{1}{a} \left[x - \frac{b}{a} \int \frac{1}{ax+b} d(ax+b) \right] + C \\ &= \frac{x}{a} - \frac{b}{a^2} \ln|ax+b| + C \end{aligned}$$

4.

拆项有理式

$$\int \frac{dx}{x(ax+b)}$$

$$\text{设 } \frac{A}{x} + \frac{B}{ax+b} = \frac{1}{x(ax+b)}$$

$$A(ax+b) + Bx = 1$$

$$Aax + Ab + Bx = 1$$

$$(Aa+B)x + Ab = 1$$

$$\begin{cases} Ab=1 \\ Aa+B=0 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{b} \\ B=-\frac{a}{b} \end{cases}$$

$$\text{原式} = \int \left[\frac{1}{bx} - \frac{a}{b(ax+b)} \right] dx$$

$$= \frac{1}{b} \int \frac{1}{x} dx - \frac{a}{b} \int \frac{1}{ax+b} d(ax+b)$$

$$= \frac{1}{b} \ln|x| - \frac{1}{b} \ln|ax+b| + C = \frac{1}{b} \left| \frac{x}{ax+b} \right| + C$$

5.

$$\begin{aligned} \int \frac{x}{(ax+b)^2} dx &= \frac{1}{a} \int \frac{ax+b-b}{(ax+b)^2} dx \\ &= \frac{1}{a} \int \left[\frac{ax+b}{(ax+b)^2} - \frac{b}{(ax+b)^2} \right] dx \\ &= \frac{1}{a} \left[\int \frac{1}{ax+b} dx - \int \frac{b}{(ax+b)^2} dx \right] \\ &= \frac{1}{a^2} \int \frac{1}{ax+b} d(ax+b) + \frac{b}{a^2} \int \frac{1}{(ax+b)^2} d(ax+b) \\ &= \frac{1}{a^2} \ln|ax+b| - \frac{b}{a^2(ax+b)} + C \end{aligned}$$

6.

$$\int \frac{1}{ax^2+b} dx$$

当 $a > 0, b > 0$ 时

$$\text{原式} = \frac{1}{\sqrt{ab}} \int \frac{1}{\sqrt{\frac{a}{b}}x^2 + \sqrt{\frac{b}{a}}} dx \quad \text{分母除以 } \sqrt{ab}$$

$$= \frac{1}{\sqrt{ab}} \int \frac{1}{\left(\sqrt{\frac{a}{b}}x\right)^2 + 1} d\left(\sqrt{\frac{a}{b}}x\right) \quad \text{分母乘 } \sqrt{\frac{a}{b}}$$

$$= \frac{1}{\sqrt{ab}} \arctan \sqrt{\frac{a}{b}}x + C$$

当 $a > 0, b < 0$ 时

$$\text{原式} = \int \left[\frac{1}{(\sqrt{a}x + \sqrt{-b})(\sqrt{a}x - \sqrt{-b})} \right] dx$$

$$\frac{A}{\sqrt{a}x - \sqrt{-b}} - \frac{B}{\sqrt{a}x + \sqrt{-b}} = \frac{1}{(\sqrt{a}x + \sqrt{-b})(\sqrt{a}x - \sqrt{-b})}$$

$$A(\sqrt{a}x + \sqrt{-b}) - B(\sqrt{a}x - \sqrt{-b}) = 1$$

$$A\sqrt{a}x + A\sqrt{-b} - B\sqrt{a}x + B\sqrt{-b} = 1$$

$$(A\sqrt{a} - B\sqrt{a})x + (A+B)\sqrt{-b} = 1$$

$$\begin{cases} A+B = -\frac{1}{\sqrt{-b}} \\ A-B = 0 \end{cases} \Rightarrow A=B = -\frac{1}{2\sqrt{-b}} = \frac{1}{2\sqrt{b}} \quad \text{★ 只有将负号放在根号中才能开根号}$$

$$\text{原式} = \frac{1}{2\sqrt{b}} \int \left(\frac{1}{\sqrt{a}x - \sqrt{-b}} - \frac{1}{\sqrt{a}x + \sqrt{-b}} \right) dx$$

$$= \frac{1}{2\sqrt{ab}} \ln|\sqrt{a}x - \sqrt{-b}| - \frac{1}{2\sqrt{ab}} \ln|\sqrt{a}x + \sqrt{-b}| + C$$

$$= \frac{1}{2\sqrt{ab}} \ln \left| \frac{\sqrt{a}x - \sqrt{-b}}{\sqrt{a}x + \sqrt{-b}} \right| + C$$

7.

$$\int \frac{x}{ax^2+b} dx = \frac{1}{2a} \int \frac{1}{ax^2+b} d(ax^2+b) \\ = \frac{1}{2a} \ln|ax^2+b| + C$$

8. 基本公式

$$\int \frac{dx}{1-x^2} = \int \frac{1}{(1+x)(1-x)} dx$$

$$\frac{A}{1+x} + \frac{B}{1-x} = \frac{1}{(1+x)(1-x)}$$

$$A - Ax + B + Bx = 1$$

$$(B-A)x + A+B = 1$$

$$\begin{cases} B-A=0 \\ A+B=1 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{2} \\ B=\frac{1}{2} \end{cases}$$

$$\text{原式} = \frac{1}{2} \left(\int \frac{1}{1+x} dx + \int \frac{1}{1-x} dx \right) \\ = \frac{1}{2} (\ln|1+x| - \ln|1-x|) + C \\ = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C$$

9.

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \int \frac{1}{1+x^2} d(x^2+1) \\ = \frac{1}{2} \ln|1+x^2| + C \\ = \frac{1}{2} \ln(1+x^2) + C \rightarrow \text{一定为正}$$

10.

$$\int \frac{1}{1+x^3} dx \quad \text{也可以背立方和公式}$$

多项式除法因式分解

$$\begin{array}{r} x^2 - x + 1 \\ x+1 \overline{) x^3 + 0x^2 + 0x + 1} \\ \underline{x^3 + x^2} \\ -x^2 + 0x + 1 \\ \underline{-x^2 - x} \\ x+1 \\ \underline{x+1} \\ 0 \end{array}$$

除式的确定方法:

代入一些简单的数(本上是
拆出x+1以便积分计算)

若 $f(a)=0$, 则 $x-a$ 一定是
 $f(x)$ 的因式

当 $a=-1$ 时, $1+(-1)^3=0$
所以 $x+1$ 是 $1+x^3$ 的因式

10. 续

$$\int \frac{1}{1+x^3} dx = \int \frac{1}{(1+x)(x^2-x+1)} dx$$

$$\frac{A}{1+x} + \frac{Bx+C}{x^2-x+1} = \frac{1}{(1+x)(x^2-x+1)} \quad \text{注意分子}$$

$$Ax^2 - Ax + A + Bx + Bx^2 + C + Cx = 1$$

$$(A+B)x^2 + (B+C-A)x + A+C = 1$$

$$\begin{cases} A+B=0 \\ B+C-A=0 \\ A+C=1 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{3} \\ B=-\frac{1}{3} \\ C=\frac{2}{3} \end{cases}$$

$$\int \frac{1}{1+x^3} dx = \frac{1}{3} \int \frac{1}{1+x} dx - \frac{1}{3} \int \frac{x-2}{x^2-x+1} dx$$

$$= \frac{1}{3} \int \frac{1}{1+x} d(1+x) - \frac{1}{3} \int \frac{x-\frac{1}{2}-\frac{3}{2}}{x^2-x+1} dx$$

$$= \frac{1}{3} \ln|1+x| - \frac{1}{3} \int \frac{x-\frac{1}{2}}{x^2-x+1} dx + \frac{1}{3} \int \frac{\frac{3}{2}}{x^2-x+1} dx$$

$$2x-1 = 2(x-\frac{1}{2})$$

$$2 \cdot \frac{1}{3} \int \frac{1}{x^2-x+1} d(x^2-x+1)$$

$$= \frac{2}{3} \ln|x^2-x+1| + C_1$$

凑基本
积分公式

$$\frac{1}{2} \int \frac{1}{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} d(x-\frac{1}{2})$$

$$= \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \arctan \frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}} + C_2$$

$$= \frac{1}{\sqrt{3}} \arctan \frac{2x-1}{\sqrt{3}} + C_2$$

$$\text{原式} = \frac{1}{3} \ln|1+x| + \frac{2}{3} \ln|x^2-x+1| + \frac{1}{\sqrt{3}} \arctan \frac{2x-1}{\sqrt{3}} + C$$

✖

11.

$$\int \frac{1}{1-x^3} dx$$

$$\begin{array}{r} -x^2 - x - 1 \\ x-1 \overline{) -x^3 + 0x^2 + 0x + 1} \\ \underline{-x^3 + x^2} \\ -x^2 + 0x \\ \underline{-x^2 + x} \\ -x + 1 \\ \underline{-x + 1} \\ 0 \end{array}$$

$$\text{原式} = \int \frac{1}{x-1} \cdot \frac{1}{x^2+x+1} dx$$

$$\frac{1}{(x-1)(x^2+x+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$$

$$Ax^2 + Ax + A + Bx^2 - Bx + Cx - C$$

$$(A+B)x^2 + (A-B+C)x + A-C = 1$$

$$\begin{cases} A-C=1 \\ A+B=0 \\ A-B+C=0 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{3} \\ B=-\frac{1}{3} \\ C=-\frac{2}{3} \end{cases}$$

$$\text{原式} = \frac{1}{3} \int \frac{1}{x-1} dx - \frac{1}{3} \int \frac{x+2}{x^2+x+1} dx$$

$$= \frac{1}{3} \int \frac{1}{x-1} d(x-1) - \frac{1}{3} \int \frac{x+\frac{1}{2}+\frac{3}{2}}{x^2+x+1} dx \quad \rightarrow 2x+1=2(x+\frac{1}{2})$$

$$= \frac{1}{3} \ln|x-1| + \frac{1}{3} \cdot \frac{1}{2} \int \frac{1}{x^2+x+1} d(\underline{x^2+x+1}) + \frac{1}{3} \cdot \frac{3}{2} \int \frac{1}{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} d(x+\frac{1}{2})$$

$$= \frac{1}{3} \ln|x-1| + \frac{1}{6} \ln|x^2+x+1| + \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \arctan \frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}} + C$$

$$= \frac{1}{3} \ln|x-1| + \frac{1}{6} \ln|x^2+x+1| + \frac{1}{\sqrt{3}} \arctan \frac{2x+1}{\sqrt{3}} + C$$

12.

$$\int \frac{1}{1+x^4} dx = \frac{1}{2} \int \frac{x^2+1-x^2+1}{1+x^4} dx \longrightarrow \text{拆项, 凑微分}$$

$$= \frac{1}{2} \int \frac{x^2+1}{1+x^4} dx - \frac{1}{2} \int \frac{x^2-1}{1+x^4} dx$$

$$= \frac{1}{2} \int \frac{1+\frac{1}{x^2}}{\frac{1}{x^2}+x^2} dx - \frac{1}{2} \int \frac{1-\frac{1}{x^2}}{\frac{1}{x^2}+x^2} dx \longrightarrow \text{上下同除 } x^2$$

$$= \frac{1}{2} \int \frac{1}{\frac{1}{x^2}+x^2} (x-\frac{1}{x}) - \frac{1}{2} \int \frac{1}{\frac{1}{x^2}+x^2} d(x+\frac{1}{x}) \longrightarrow \text{凑微分}$$

$$= \frac{1}{2} \int \frac{1}{(x-\frac{1}{x})^2+2} d(x-\frac{1}{x}) - \frac{1}{2} \int \frac{1}{(x+\frac{1}{x})^2+2} d(x+\frac{1}{x}) \longrightarrow \text{凑基本积分公式}$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \arctan \frac{x-\frac{1}{x}}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{2\sqrt{2}} \ln \left| \frac{x+\frac{1}{x}-\sqrt{2}}{x+\frac{1}{x}+\sqrt{2}} \right| + C \quad \text{注意公式:}$$

$$= \frac{\sqrt{2}}{4} \arctan \frac{\sqrt{2}(x^2-1)}{2x} - \frac{1}{4\sqrt{2}} \ln \left| \frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x-1} \right| + C \quad \int \frac{1}{x^2+a^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C$$

不要记反了!!!

13.

$$\int \frac{dx}{1-x^4} = \int \frac{1}{(1+x^2)(1-x^2)} dx$$

$$= \frac{1}{2} \int \left(\frac{1}{1+x^2} + \frac{1}{1-x^2} \right) dx$$

$$= \frac{1}{2} \int \frac{1}{1+x^2} dx + \frac{1}{2} \int \frac{1}{1-x^2} dx$$

$$= \frac{1}{2} \arctan x + \frac{1}{2} \cdot \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C$$

$$= \frac{1}{2} \arctan x + \frac{1}{4} \ln \left| \frac{1+x}{1-x} \right| + C$$

$$\text{设 } \frac{Ax+B}{1+x^2} + \frac{Cx+D}{1-x^2} = \frac{1}{(1+x^2)(1-x^2)}$$

$$(Ax+B)(1-x^2) + (Cx+D)(1+x^2)$$

$$= Ax - Ax^3 + B - Bx^2 + Cx + Cx^3 + D + Dx^2$$

$$= (C-A)x^3 + (D-B)x^2 + (A+C)x + B+D$$

$$\begin{cases} C-A=0 \\ D-B=0 \\ A+C=0 \\ B+D=1 \end{cases}$$

$$\Rightarrow \begin{cases} A=C=0 \\ B=D=\frac{1}{2} \end{cases}$$

$$\frac{1}{(x^2+1)^2} dx$$

$$\frac{1}{2} \int \frac{x^4+1-x^4+1}{1+x^4} dx = \frac{1}{2} \int \frac{1+x^4}{1+x^4} dx - \frac{1}{2} \int \frac{x^4-1}{1+x^4}$$

$$\int \frac{1}{1+x^4} dx = \frac{1}{2} \int 1 dx - \frac{1}{2} \int \frac{x^4}{1+x^4} dx + \frac{1}{2} \int \frac{1}{1+x^4} dx$$

$$\boxed{\frac{1}{2} \int \frac{1}{1+x^4} dx} = \frac{1}{2} x - \frac{1}{2} \int \frac{x^4+1-1}{1+x^4} dx$$

$$= \frac{1}{2} x - \frac{1}{2} \int \frac{x^4+1}{1+x^4} dx \quad \boxed{\frac{1}{2} \int \frac{1}{1+x^4} dx}$$

減沒了!!!

錯誤

14. 例 10.11

$$\int \frac{x}{1+x^3} dx$$

$$\begin{array}{r} x^2 - x + 1 \\ x+1 \overline{) x^3 + 0x^2 + 0x + 1} \\ \underline{x^3 + x^2} \\ -x^2 + 0x + 1 \\ \underline{-x^2 - x} \\ x + 1 \\ \underline{x + 1} \\ 0 \end{array}$$

设 $\frac{A}{x+1} + \frac{Bx+C}{x^2-x+1} = \frac{x}{(x+1)(x^2-x+1)}$

$$Ax^2 - Ax + A + Bx^2 + Bx + Cx + C = x$$

$$(A+B)x^2 + (B-A+C)x + A+C = x$$

$$\begin{cases} A+B=0 \\ B-A+C=1 \\ A+C=0 \end{cases} \Rightarrow \begin{cases} A=-\frac{1}{3} \\ B=\frac{1}{3} \\ C=\frac{1}{3} \end{cases}$$

$$\text{原式} = \int \frac{x}{(x+1)(x^2-x+1)} dx$$

$$= -\frac{1}{3} \int \frac{1}{x+1} dx + \frac{1}{3} \int \frac{x+1}{x^2-x+1} dx$$

$$= -\frac{1}{3} \ln|x+1| + \frac{1}{3} \int \frac{x-\frac{1}{2}+\frac{3}{2}}{x^2-x+1} dx$$

$$= -\frac{1}{3} \ln|x+1| + \frac{1}{3} \cdot \frac{1}{2} \int \frac{1}{x^2-x+1} d(x^2-x+1) + \frac{1}{3} \cdot \frac{3}{2} \int \frac{1}{(x-\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} d(x-\frac{1}{2})$$

$$= -\frac{1}{3} \ln|x+1| + \frac{1}{6} \ln|x^2-x+1| + \frac{1}{\sqrt{3}} \arctan \frac{2x-1}{\sqrt{3}} + C$$

15.

$$\int \frac{x}{1-x^3} dx$$

$$\begin{array}{r} x-1 \overline{) \begin{array}{r} -x^2-x-1 \\ -x^3+0x^2+0x+1 \\ \hline -x^2+0x+1 \\ -x^2+x \\ \hline -x+1 \\ -x+1 \\ \hline 0 \end{array}} \end{array}$$

设 $\frac{A}{x-1} + \frac{Bx+C}{x^2+x+1} = \frac{x}{(x-1)(x^2+x+1)}$

$$Ax^2+Ax+A+Bx^2+Cx-Bx-C=x$$

$$(A+B)x^2+(A+C-B)x+A-C=x$$

$$\begin{cases} A+B=0 \\ A+C-B=1 \\ A-C=0 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{3} \\ B=-\frac{1}{3} \\ C=\frac{1}{3} \end{cases}$$

$$\text{原式} = -\int \frac{x}{(x-1)(x^2+x+1)} dx$$

$$= -\frac{1}{3} \int \frac{1}{x-1} dx - \int \frac{-\frac{1}{3}x + \frac{1}{3}}{x^2+x+1} dx$$

$$= -\frac{1}{3} \int \frac{1}{x-1} d(x-1) + \frac{1}{3} \int \frac{x-1}{x^2+x+1} dx$$

$$= -\frac{1}{3} \ln|x-1| + \frac{1}{3} \int \frac{x+\frac{1}{2}-\frac{3}{2}}{x^2+x+1} dx$$

$$= -\frac{1}{3} \ln|x-1| + \frac{1}{3} \int \frac{x+\frac{1}{2}}{x^2+x+1} dx - \frac{1}{3} \cdot \frac{3}{2} \int \frac{1}{x^2+x+1} dx$$

$$= -\frac{1}{3} \ln|x-1| + \frac{1}{3} \cdot \frac{1}{2} \int \frac{1}{x^2+x+1} d(x^2+x+1) - \frac{1}{2} \int \frac{1}{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} dx$$

$$= -\frac{1}{3} \ln|x-1| + \frac{1}{6} \ln|x^2+x+1| - \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \arctan \frac{\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}}} + C$$

$$= -\frac{1}{3} \ln|x-1| + \frac{1}{6} \ln|x^2+x+1| - \frac{1}{\sqrt{3}} \arctan \frac{2x+1}{\sqrt{3}} + C$$

16.

$$\int \frac{x}{1+x^4} dx = \frac{1}{2} \int \frac{1}{1+x^2} d(x^2+1) \\ = \frac{1}{2} \arctan x^2 + C$$

17.

$$\int \frac{x}{1-x^4} dx = \frac{1}{2} \int \frac{1}{1^2 - (x^2)^2} d(x^2) \\ = \frac{1}{2} \cdot \frac{1}{2} \ln \left| \frac{x^2+1}{x^2-1} \right| \\ = \frac{1}{4} \ln \left| \frac{x^2+1}{x^2-1} \right| + C$$

18.

$$\int \frac{dx}{c^2+x^2} = \frac{1}{c} \arctan \frac{x}{c} + C$$

19.

$$\int \frac{dx}{(c^2+x^2)^2} = -\frac{1}{2} \int \frac{1}{x} d\left(\frac{1}{c^2+x^2}\right)$$

$$= -\frac{1}{2} \left(\frac{1}{x(c^2+x^2)} - \int \frac{1}{c^2+x^2} d\left(\frac{1}{x}\right) \right)$$

$$= -\frac{1}{2} \left(\frac{1}{x(c^2+x^2)} + \int \frac{dx}{x^2(c^2+x^2)} \right)$$

$$= -\frac{1}{2x(c^2+x^2)} - \frac{1}{2} \int \left(\frac{1}{c^2x^2} - \frac{1}{c^2(c^2+x^2)} \right) dx$$

$$= -\frac{1}{2x(c^2+x^2)} - \frac{1}{2c^2} \left(\int \frac{1}{x^2} dx + \int \frac{1}{c^2+x^2} dx \right)$$

$$= -\frac{1}{2x(c^2+x^2)} + \frac{1}{2c^2} \left(-\frac{1}{x} + \frac{1}{c} \arctan \frac{x}{c} \right) + C_1$$

$$\frac{Ax+B}{x^2} + \frac{Dx+F}{x^2+c^2} = \frac{1}{x^2(c^2+x^2)}$$

$$(Ax+B)(x^2+c^2) + x^2(Dx+F) = 1$$

$$= Ax^3 + Ac^2x + Bx^2 + Bc^2 + Dx^3 + Fx^2$$

$$\begin{cases} A+D=0 \\ Ac^2=0 \\ B+F=0 \\ Bc^2=1 \end{cases} \Rightarrow \begin{cases} A=0 \\ B=\frac{1}{c^2} \\ D=0 \\ F=-\frac{1}{c^2} \end{cases}$$

20.

$$\int \frac{1}{(c^2+x^2)^3} dx \rightarrow \left(\frac{1}{(c^2+x^2)^2} \right)' = \frac{-2(c^2+x^2) \cdot 2x}{(c^2+x^2)^4} = \frac{-4x}{(c^2+x^2)^3}$$

$$= -\frac{1}{4} \int \frac{1}{x} d \frac{1}{(c^2+x^2)^2} \quad \text{处理方法同19}$$

$$= -\frac{1}{4} \left[\frac{1}{x(c^2+x^2)^2} - \int \frac{1}{(c^2+x^2)^2} d \frac{1}{x} \right]$$

$$= -\frac{1}{4} \left[\frac{1}{x(c^2+x^2)^2} + \int \frac{1}{x^2(c^2+x^2)^2} dx \right]$$

$$= -\frac{1}{4} \left\{ \frac{1}{x(c^2+x^2)^2} + \int \left[\frac{1}{c^4 x^2} - \frac{1}{c^4(c^2+x^2)} + \frac{1}{c^2(c^2+x^2)^2} \right] dx \right\} \quad \textcircled{1}$$

$$= -\frac{1}{4x(c^2+x^2)^2} - \frac{1}{4} \left[\frac{1}{c^4} \int \frac{1}{x^2} dx - \frac{1}{c^4} \int \frac{1}{c^2+x^2} dx + \frac{1}{c^2} \int \frac{1}{(c^2+x^2)^2} dx \right]$$

$$= -\frac{1}{4x(c^2+x^2)^2} - \frac{1}{4c^4} \cdot \left(-\frac{1}{x}\right) + \frac{1}{4c^4} \cdot \frac{1}{c} \arctan \frac{x}{c} - \frac{1}{4c^2} \cdot \frac{1}{2} \int \frac{1}{x} d \frac{1}{c^2+x^2}$$

$$= -\frac{1}{4x(c^2+x^2)^2} + \frac{1}{4c^4 x} + \frac{1}{4c^5} \arctan \frac{x}{c} - \frac{1}{8c^2} \left(\frac{1}{x(c^2+x^2)} - \int \frac{1}{c^2 x^2} dx - \int \frac{1}{c^2(c^2+x^2)} dx \right)$$

$$= -\frac{1}{4x(c^2+x^2)^2} + \frac{1}{4c^4 x} + \frac{1}{4c^5} \arctan \frac{x}{c} - \frac{1}{8c^2} \left[\frac{1}{x(c^2+x^2)} - \frac{1}{c^2} \left(-\frac{1}{x}\right) - \frac{1}{c^2} \cdot \frac{1}{c} \arctan \frac{x}{c} \right] + C$$

$$= -\frac{1}{4x(c^2+x^2)^2} + \frac{1}{4c^4 x} + \frac{1}{4c^5} \arctan \frac{x}{c} - \frac{1}{8c^2 x(c^2+x^2)} - \frac{1}{8c^4 x} + \frac{1}{8c^5} \arctan \frac{x}{c} + C$$

①

$$\frac{Ax+B}{x^2} + \frac{Dx+F}{c^2+x^2} + \frac{Gx+H}{(c^2+x^2)^2} = \frac{1}{x^2(c^2+x^2)^2}$$

$$(Ax+B)(c^2+x^2)^2 + (Dx+F)x^2(c^2+x^2) + (Gx+H)x^2$$

$$= (Ax+B)(c^4+2c^2x^2+x^4) + (Dx+F)(x^2c^2+x^4) + (Gx^3+Hx^2)$$

$$= Ac^4x + 2Ac^2x^3 + Ax^5 + Bc^4 + 2Bc^2x^2 + Bx^4 + Dc^2x^3 + Dx^5 + Fc^2x^2 + Fx^4 + Gx^3 + Hx^2$$

$$= (A+D)x^5 + (B+F)x^4 + (2Ac^2+Dc^2+G)x^3 + (2Bc^2+Fc^2+H)x^2 + Ac^4x + Bc^4$$

$$\begin{cases} A+D=0 \\ B+F=0 \\ 2Ac^2+Dc^2+G=0 \\ 2Bc^2+Fc^2+H=0 \\ Ac^4=0 \\ Bc^4=1 \end{cases} \Rightarrow$$

$$\begin{cases} A=D=0 \\ B=\frac{1}{c^4} \\ F=-\frac{1}{c^4} \\ G=0 \\ H=\frac{1}{c^2} \end{cases}$$

21.

$$\int \frac{x}{c^2+x^2} dx = \frac{1}{2} \int \frac{1}{c^2+x^2} d(x^2+c^2)$$

$$= \frac{1}{2} \ln(c^2+x^2) + C_1$$

22.

$$\int \frac{x}{(c^2+x^2)^2} dx = \frac{1}{2} \int \frac{1}{(c^2+x^2)^2} d(x^2+c^2)$$

$$= -\frac{1}{2(c^2+x^2)} + C_1$$

23.

$$\int \frac{x^2}{c^2+x^2} dx = \int \frac{x^2+c^2-c^2}{c^2+x^2} dx$$

$$= \int 1 dx - c^2 \int \frac{1}{c^2+x^2} dx$$

$$= x - c^2 \cdot \frac{1}{c} \arctan \frac{x}{c} + C_1$$

$$= x - c \arctan \frac{x}{c} + C_1$$

24.

$$\int \frac{x^2}{(c^2+x^2)^2} dx = \int \frac{x^2+c^2-c^2}{(c^2+x^2)^2} dx$$

$$= \int \frac{1}{c^2+x^2} dx - c^2 \int \frac{1}{(c^2+x^2)^2} dx$$

$$= \frac{1}{c} \arctan \frac{x}{c} + \frac{c^2}{2} \int \frac{1}{x} d\left(\frac{1}{c^2+x^2}\right)$$

$$= \frac{1}{c} \arctan \frac{x}{c} + \frac{c^2}{2} \left(\frac{1}{x(c^2+x^2)} + \int \frac{1}{x^2(c^2+x^2)} dx \right)$$

$$= \frac{1}{c} \arctan \frac{x}{c} + \frac{c^2}{2} \left(\frac{1}{x(c^2+x^2)} + \int \frac{1}{c^2 x^2} dx - \frac{1}{c^2} \int \frac{1}{c^2+x^2} dx \right)$$

$$= \frac{1}{c} \arctan \frac{x}{c} + \frac{c^2}{2} \left(\frac{1}{x(c^2+x^2)} - \frac{1}{c^2} \cdot \frac{1}{x} - \frac{1}{c^2} \cdot \frac{1}{c} \arctan \frac{x}{c} \right) + C_1$$

$$= \frac{1}{c} \arctan \frac{x}{c} + \frac{c^2}{2x(c^2+x^2)} - \frac{1}{2x} - \frac{1}{2c} \arctan \frac{x}{c} + C_1$$

$$= \frac{1}{2c} \arctan \frac{x}{c} + \frac{c^2}{2x(c^2+x^2)} - \frac{1}{2x} + C_1$$

$$= \frac{1}{2c} \arctan \frac{x}{c} - \frac{x}{2(c^2+x^2)} + C_1$$

$$\frac{1}{x^2(c^2+x^2)} = \frac{A}{x^2} + \frac{B}{c^2+x^2}$$

$$= \frac{A(x^2+c^2) + Bx^2}{x^2(c^2+x^2)}$$

$$Ax^2 + Ac^2 + Bx^2 = 1$$

$$\begin{cases} A+B=0 \\ AC^2=1 \end{cases} \Rightarrow \begin{cases} A = \frac{1}{c^2} \\ B = -\frac{1}{c^2} \end{cases}$$